

Vectors in $\mathbb{R}^3$

$$+ : \mathcal{V}_3 \times \mathcal{V}_3 \rightarrow \mathcal{V}_3$$

$$(\vec{u}, \vec{v}) \mapsto \alpha \vec{u} + \vec{v}$$

$(\mathcal{V}_3, +)$ is a com. group

$$\vec{a} \cdot \vec{b} \leq 0 \Rightarrow \vec{a} \perp \vec{b}$$

$$\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$$

$$\vec{b} = b_1 \vec{i} + b_2 \vec{j} + b_3 \vec{k}$$

$$\vec{c} = c_1 \vec{i} + c_2 \vec{j} + c_3 \vec{k}$$

$$(\vec{a}, \vec{b}, \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$\Delta = 0 \Leftrightarrow$ the vectors lie in the same plane

$(\vec{a}, \vec{b}, \vec{c})$ have pos. orientation if $\Delta > 0$
neg. orientation if $\Delta < 0$

$\vec{a} \times \vec{b}$ has pos. orientation as $\{\vec{i}, \vec{j}, \vec{k}\}$

$$\text{ex. } \mathbb{R}^3: \vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}$$

$$\vec{v} = \vec{i} - \vec{j} + \vec{k}$$

$$\vec{a} \parallel \vec{v}_1$$

$$\vec{a} \perp \vec{v}_2 \Rightarrow$$

~~$$\vec{a} \cdot \vec{b} = 0 \Rightarrow$$~~

$$\vec{v}_1 = \alpha \vec{a}$$

$$\vec{v}_2 = \alpha \vec{a} +$$

$$\vec{v} - \alpha \vec{a}$$

$$V - \alpha \vec{A} \perp \vec{a} \Rightarrow \vec{a} \cdot (\vec{V} - \alpha \vec{A}) = 0$$

$$\vec{a} \cdot \vec{V} - \alpha \vec{a} \cdot \vec{a} = 0$$

$$\alpha = \frac{\vec{V} \cdot \vec{a}}{\vec{a} \cdot \vec{a}}$$